

Shahid Mohammad Mulla

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Education

•University of California, San Diego

Sept 2025 - June 2027

MS in ECE (Major: Intelligent Systems, Robotics and Controls)

•Indian Institute of Technology, Madras

Nov 2021 - Jul 2025

Bachelor of Technology in Mechanical Engineering

Professional Experience

•Ather Energy (India's leading premium two-wheeled EV company)

Jun 25'-Sept 25'

Controls Engineer

Systems Intelligence Lab

- Explored **Multi-Rate Controller** using **discrete LQR** on the **BMS model** for **CV charging** to optimize the controller.
- Designed an **optimized charging cycle** to achieve the most accurate **State of Health (SoH)** estimation of the battery.
- Developed a **state estimation framework** using an **Intermittent Kalman Filter** with **Unknown Input Observer** for cloud-based monitoring, reducing live scooter data transmission by **88%** while preserving **accuracy**. (Patent filed)

Publications

[1] **Shahid Mohammad Mulla**, Aryan K., Lakshmi Narasimhan, Anuj K. Tiwari - "RF-Source Seeking with Obstacle Avoidance using Real-time Modified Artificial Potential Fields in Unknown Environments" oral presentation at [ICCAS 2025](#), Incheon (South Korea). ([Arxiv Preprint](#))

Research Experience

•Sim-to-Real Transfer for Dexterous Bi-Manipulation of Articulated Tools

May 26'-Present

Graduate Student Researcher

[ARC Lab](#), UCSD

- Implementing a **teacher-student framework** targeting zero-shot **sim-to-real** transfer for **dexterous manipulation**.
- Architected the full **NVIDIA Isaac Sim** environment — **MDP**, **action/observation spaces**, and **reward shaping**.
- Developed density-based **curriculum learning** with a **safety curriculum** for **collision-aware grasping** in RL training.
- Designed **structured reward landscapes** to train stable RL policies for **multi-fingered dexterous manipulation**.

•ADMM-Based Receding-Horizon Control for Contact-Rich Non-Prehensile Manipulation

April 26'-Present

Graduate Student Researcher

[ERL](#), UCSD

- Modelled a **two-body planar system** (box + circle) under two convex contact models, **Anitescu QP** and **ContactSDF**.
- Implemented an **ADMM-based receding-horizon controller** over both **Anitescu QP** and **ContactSDF** dynamics, achieving real-time **contact-aware trajectory optimisation** with **obstacle avoidance** and **workspace constraints**.
- Extending the framework to **6-joint robotic arm manipulators** and deploying it on **xArm6** for real-world validation.

•Object-Informed MPPI Control for Non-Prehensile Manipulation

Nov 25'-March 26'

Graduate Student Researcher

[ERL](#), UCSD

- Formulated a **hierarchical MPPI** decomposing non-prehensile manipulation into decoupled **object-level** and **robot-level** optimal control problems via **contact-implicit** complementarity dynamics.
- Designed **SOI** & **CLOI** with open/closed-loop object replanning, boosting task success by **40%** over vanilla MPPI.
- Deployed GPU-parallelized rollouts in **NVIDIA Isaac Gym** on a **6-DoF xArm6** using **ROS2** for communication.
- Integrated **SAM6D** for **6-DoF object pose estimation** via **Intel RealSense** for closed-loop state feedback on hardware.

Distinguished Projects

•SmolVAM: Generalizable Robot Control with Small Video-Action Models

April 26' - Jun 26'

Robot Learning & Manipulation

- Decoupled world modeling from action inference using **0.7B Hummingbird** to synthesize future visual rollouts.
- Adapted Hummingbird to robotic manipulation via **LoRA fine-tuning**, training only **~0.1%** of parameters.
- Trained a **Flow Matching** action decoder on **VideoMAE** spatiotemporal tokens for action chunk prediction.
- Achieved **60% success rate** on Cube Stack task in MuJoCo with full pipeline being implemented on **LeRobot**.

•Design, Fabrication & Control of an Autonomous Underwater Vehicle (AUV)

Sept 23'-Nov 24'

Undergraduate Student Researcher

[MAV Lab](#), IIT Madras

- Modelled the **dynamics** of the AUV in ROS2 to create a simulation of our AUV & visualized it in Gazebo.
- Used **DVL**, **IMU** & **depth sensor** for **localization** and used a **stereo camera** in the front & a **monocamera** at the bottom for **object detection & odometry**. Fused the sensor data using **Extended Kalman Filter** for **SLAM**.
- Implemented **Line of Sight Guidance Control** with **waypoint navigation** if the desired location has been computed.
- Implemented **obstacle & gate detection** using **YOLOv5** model & fine tuned it with custom dataset on **Roboflow**.

Technical Skills

Areas: Robotics, SLAM, Manipulation, VLA/VLMs, Quadrupeds, Humanoids, Controls, RL, Motion Planning, Computer Vision, AI, Deep Learning, ML, Sensor Fusion, Optimization, Linux, Multi Threading, Real-Time Systems

Programming Languages: MATLAB, Python, C, C++

Tools & Frameworks: ROS2, Simulink, OpenCV, Open3D, PyTorch, Scikit-learn, Isaac Lab/Sim, Docker, Git